Object Oriented Programming Lab

SPRING - 2024

LAB 05

FAST National University of
Computer and Emerging Sciences

Learning Outcomes

In this lab you are expected to learn the following:

- Structures
- Recursion

Structures

Structure is a user-defined data type that allows you to combine data items of different kinds.

Structure Declaration

```
struct Books {
     int book_id;
     int pages;
     int price;
};
```

Types of Members

Structures in C++ can contain two types of members:

- **Data Member:** These members are normal C++ variables. We can create a structure with variables of different data types in C++.
- **Member Functions:** These members are normal C++ functions. Along with variables, we can also include functions inside a structure declaration.

Accessing Structure Members

```
#include<iostream>
using namespace std;
struct Books {
     int book_id;
     int pages;
     int price;
};
int main() {
Books Book1;
Books Book2;
Book1.book_id=123;
cin>>Book1.pages;
cin>>Book2.pages;
}
```

Array of Structures

```cpp
#include<iostream>
using namespace std;
struct Books {
    int book_id;
    int pages;
    int price;
};
int main() {
    Book b1[3]={{1,275,70},{2,600,90},{3,786,100}};
    //can also assigned values using cin
    for(int i=0; i<3; i++){
        cin>>b1[i].book_id;
        cin>>b1[i].pages;
        cin>>b1[i].price;
    }
}
```

RECURSION

A recursive function is one that calls itself.

Example

```cpp
void print(int n) {

    if ( n <= 0 )

    return; //Base condition

    cout << n << " "; //Prints number n

    print(n-1); //Calls itself with (n-1)

    return; //Returns from the function

}
```

Output

```
5 4 3 2 1
C:\Users\hp\source\repos\Lab01\x64\Debug\Lab01.exe (process 4736) exited with code 0.
Press any key to close this window . . .
```

Lab Tasks

Submission Instructions:

1. Create a single cpp file containing all the functions of the problems and main function.
2. Save the **cpp** file with the task number
   e.g. Q1.cpp
3. Now create a new folder with name *ROLLNO_SEC_LAB01* **e.g. i22XXXX_A_LAB06**
4. You need to display your roll no and name before the output of each question.
5. Move all of your .**cpp files (without the main function i.e., comment out the mainfunction)** to this newly created directory and compress it into a **.zip file**.
6. Now you have to submit this zipped file on Google Classroom.
7. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**
8. Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Question 1.

A. Declare a structure named Complex having two data members named;
   - real of type int
   - imaginary of type int.

B. Create a member function addComplex that takes two arguments, the first argument should be passed by value and the second should be passed by reference. Your addComplex function should add two complex numbers using structures. To add two complex numbers, just add the corresponding real and imaginary parts. For Example, the sum of 5 + 3*i* and 4 + 2*i* is 9 + 5*i*.

Prototype:
Complex addComplex(Complex c1, Complex &c2);

Question 2.

A. Declare a structure Named Student to store the information of 5 students taken from the user. The Data members should be:
   - Name of type string
   - roll_no of type string
   - age of type int

B. Write a member function getNames which takes a dynamic array of type Student of size 5 and will return the names of all students having age 18.

Prototype:
string* getNames(Student *std);

C. Write a member function getEvenRollno which takes a dynamic array of type Student of size 5 and will return information of all students having even roll numbers. In order to get an even roll number just check its last digit.

Prototype:
Student* getEvenRollno(Student *std);

Question 3.

A. Declare a structure named as CustomTime having three data members named;
   - hours of type int
   - mins of type int
   - secs of type int

B. Write a member function timeToSecs which takes CustomTime object t1 as a function parameter, Calculate the total seconds in time and convert this object t1 to seconds (of type int) and return it.

Prototype:
int timeToSecs(CustomTime t1);

C. Write a member function AddTimes which takes CustomTime object t1 and CustomTime object t2 as parameters.
   1. You have to convert these two objects t1, t2 in seconds using above-defined function timeToSecs.
   2. Add both seconds returned from above step.

Prototype:
int AddTimes(CustomTime t1, CustomTime t2);

Question 4.

A. Declare a structure named as CourseRegistration having data members named;
   - courseCode of type string
   - courseTitle of type string
   - CreditHours of type int
   - Section of type char

B. Declare a structure named as SemesterRegistration having data members named;
   - semesterCode of type string
   - course_reg of type CourseRegistration (a dynamic array of size 5)

C. Write a member function GetCreditHoursCount which takes SemesterRegistration object s as parameter and return total numbers of credit hours registered in it.

Prototype:
int GetCreditHoursCount(semesterRegistration s);

D. Write a member function FindCourseInSemesterRegistration which takes SemesterRegistration object s and a course code as parameter and return true if the course is registered in the semester.

Prototype:

bool FindCourseInSemesterRegistration(semesterRegistration s, string courseCode)

Question 5.

Print all numbers in range [100,999] such that sum of digits at even position is equal to sum of digits at odd position

Output:

110 121 132 143 154 165 176 187 198 220 231 242 253 264 275 286 297 330 341 352 363 374 385 396 440 451 462 473 484 495 550 561 572 583 594 660 671 682 693 770 781 792 880 891 990